

SPIM Excitation Lightpath Design with ABCD Matrix

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General light path design

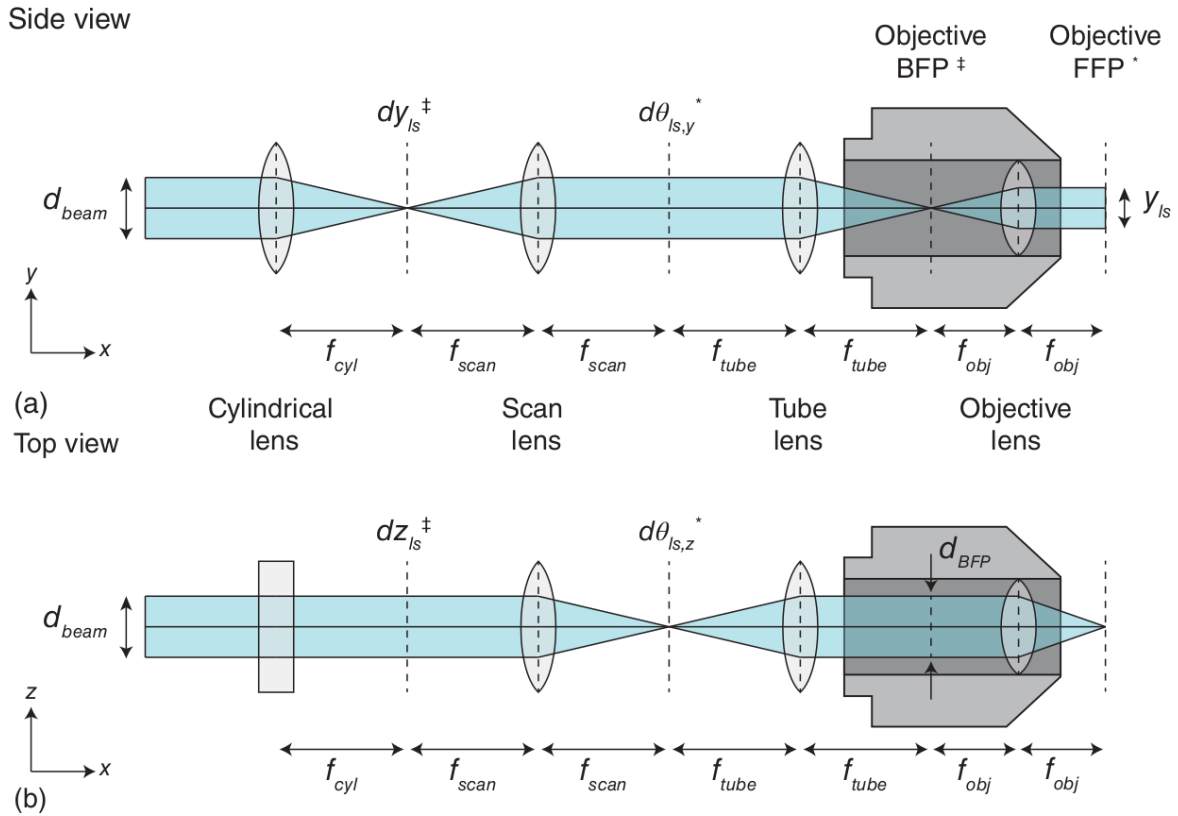


Figure 1: Relayed generation of a well-corrected static light sheet (*Huisken et al., 2004*)

Optical elements

Symbol	Parameters	Description	Part #
L_{cyl}	$f_{cyl} = 75mm$	Cylindrical lens	ThorLabs LJ1703RM-A
L_{scan}	$f_{scan} = 50mm$	Scanning lens	ThorLabs AC254-050-A
L_{tube}	$f_{tube} = 30mm$	Tube lens	ThorLabs AC254-030-A
L_{obj}	$f_{obj} = 45mm$	Objective 4X	Olympus PLT 4X, NA 0.1

Element matrices

$$M = \begin{bmatrix} 1 & z_{obj} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1/f_{obj} & 1 \end{bmatrix} \begin{bmatrix} 1 & f_{obj} + f_{tube} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1/f_{tube} & 1 \end{bmatrix} \begin{bmatrix} 1 & f_{tube} + f_{scan} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1/f_{scan} & 1 \end{bmatrix} \begin{bmatrix} 1 & f_{scan} + f_{cyl} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1/f_{cyl} & 1 \end{bmatrix} \begin{bmatrix} 1 & z_0 \\ 0 & 1 \end{bmatrix} \quad (1)$$

$$y_0 \geq d_{beam}/2 \quad (2)$$

Imaging chamber configuration